Case 40: Wind–slope vector and direction sweep

1 Verification purpose

This coupling case verifies how ELMFIRE combines Rothermel wind and slope factors into a direction of maximum spread (DMS), head speed, and fire ellipse. Wind direction rotates through 0, 45, 90, 135, and 180 degrees relative to upslope while magnitudes remain fixed. It can expose degree/radian errors, wind-from/wind-to reversal, aspect reversal, swapped raster axes, scalar addition in place of vector addition, or incorrect ellipse construction. The question is whether all simulation conditions meet the DMS, head rate of spread (ROS), length/width, and intensity limits.

2 Mathematical formulation and numerical implementation

Let upslope azimuth be zero and wind-to azimuth be α. The dimensionless forcing vector is

$$\boldsymbol{\phi} = \phi_s(0, 1) + \phi_w(\sin\alpha, \cos\alpha), \tag{1}$$

so $\theta_{DMS} = \text{atan2}(\phi_x, \phi_y)$ and $R_H = R_0(1 + |\boldsymbol{\phi}|)$. ELMFIRE derives an effective wind from $|\boldsymbol{\phi}|$ and uses its Rothermel length-to-width relation, capped at eight. Because these quantities are defined on the sloping surface, the preprocessor makes the initial circle and the postprocessor measures distances in slope-surface coordinates rather than mixing them with map-plane northing. The scalar factors follow Andrews (2018), RMRS-GTR-371; vectoring and ellipse propagation exercise ELMFIRE's level-set implementation.

3 Derivation of expected behavior

Aligned wind and slope add; opposed forcing subtracts in vector space and the stronger component sets direction. At intermediate angles, the expected DMS lies between the two forcing directions, not generally at the arithmetic mean of angles. The case-specific reference calculation independently evaluates ϕ_s, ϕ_w, their vector magnitude and azimuth, head ROS, effective wind, and length/width.

Let upslope define zero azimuth and let α be the generated wind-to-upslope angle. With $\phi_s = 6.436561$, $\phi_w = 9.746235$, and $R_0 = 0.9178563\ \text{m min}^{-1}$, the vector reference is

$$\phi_x = \phi_s + \phi_w \cos\alpha, \quad \phi_y = \phi_w \sin\alpha, \quad \phi = \sqrt{\phi_x^2 + \phi_y^2}, \quad \theta = \text{atan2}(\phi_y, \phi_x), \quad R_h = R_0(1 + \phi).$$

For example, at $\alpha = 90$ degrees,

$$\phi = \sqrt{6.436561^2 + 9.746235^2} = 11.679829,$$

$$\theta = \tan^{-1}(9.746235/6.436561) = 56.5586 \text{ degrees},$$

and

$$R_h = 0.9178563(1 + 11.679829) = 11.638261\ \text{m min}^{-1}.$$

The effective vector wind is 4.41571 mph $= 1.97394\ \text{m s}^{-1}$; substitution in $L/W = 0.936e^{0.2566U} + 0.461e^{-0.1548U} - 0.397$ gives $L/W = 1.495859$. Across $\alpha = \{0, 45, 90, 135, 180\}$ degrees, the reference $(\theta, R_h, L/W)$ rows are

$$(0, 15.7713, 1.63745), \quad (27.3422, 14.6898, 1.60074),$$

$$(56.5586, 11.6383, 1.49586), \quad (93.7779, 7.25716, 1.33838),$$

$$(180, 3.95566, 1.20547).$$

Direction error is circular, $e_\theta = \min(|\theta_m - \theta_r|, 360 - |\theta_m - \theta_r|)$. The other metrics are relative errors in R_h, L/W, and $I_R = 686.5297$ kW m^{-2}; their maximum substitutions are $e_\theta \leq 5$ degrees, $e_R \leq 0.05$, $e_{LW} \leq 0.10$, and $e_I \leq 0.01$.

4 Verification metrics and acceptance criteria

Maximum circular DMS error is 5°, maximum head-ROS relative error is 0.05, maximum length/width relative error is 0.10, and maximum intensity relative error is 0.01. DMS is inferred from the farthest one percent of the burned footprint, subject to at least 100 burned cells and at least five farthest-front samples. Head ROS is a time of arrival (TOA) regression in a narrow predicted-axis corridor with at least ten samples. Length/width uses full-footprint slope-surface extents projected along and across the DMS. Reaction intensity is the median over valid burned cells and requires at least 90 percent coverage of that footprint. These complementary metrics distinguish direction, speed, shape, and output completeness. Every simulation condition is required.

5 Simulation configuration and predicted outcome

Fuel model 2 is used on a 1400 m square, 10 m grid with a 30 m-radius signed-distance ignition clipped to $[-1, 1]$. Slope is requested as 50% and generated as the integer 27-degree value used by ELMFIRE (50.95% effective rise/run), with upslope north; target midflame wind is 4 mph. Dead moistures are 5/7/9%, live-herb moisture is 60%, and live-woody moisture is 90% in both every generated namelist and the independent reference calculation. Wind-from azimuth is generated so the wind-to vector has the required relative angle. Each simulation condition uses the largest conservative whole-second timestep no greater than the 5%-of-cell target that divides the fixed 1800 s duration exactly. This retains Courant–Friedrichs–Lewy (CFL) margin and prevents a shortened final step from triggering ELMFIRE's stalled-front timestamp defect. Crown fire, spotting, WUI, stochastic wind, and acceleration are disabled. The footprint should rotate smoothly and change head extension as the vectors move from aligned to opposed. The evaluator accepts only unique TOA and reaction-intensity rasters identified by a final dump record. A stalled-final record is accepted only when subtracting the configured meteorology interval reconstructs the 1800 s stop time within one configured timestep.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Whole-footprint direction and shape
Fuel/slope	FM2; 50	code / percent	Fixed slope forcing
Midflame wind	4	mph	Fixed wind-factor magnitude
Relative angles	0, 45, 90, 135, and 180 degrees	degrees	Vector-coupling sweep
Time control	generated whole-second divisor; stop 1800	s	Exact timestep multiple

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

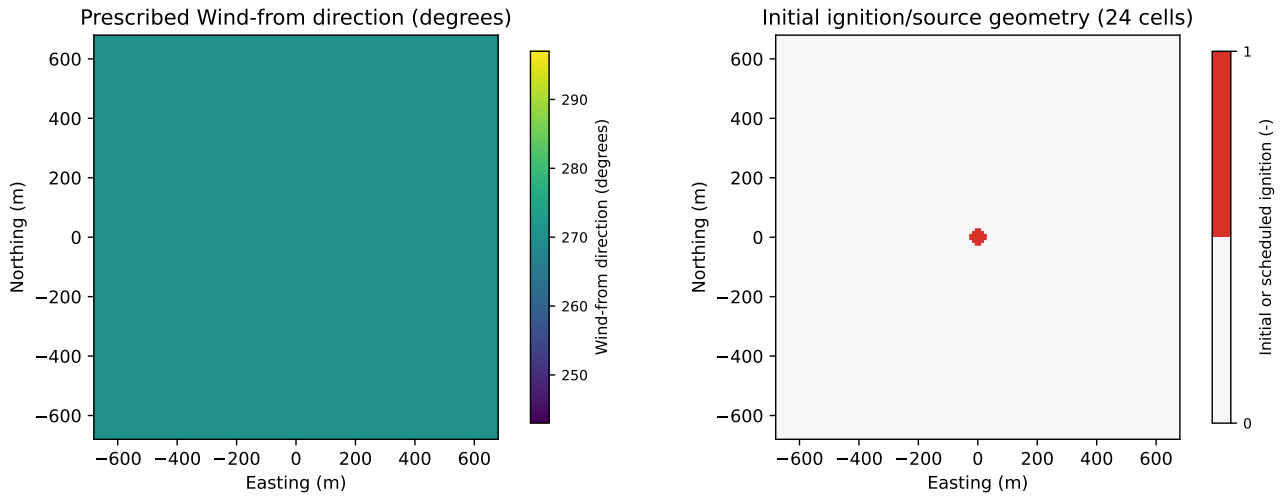


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 5 of 5 simulation conditions and reports **PASS**. The generated diagnostic is: —.

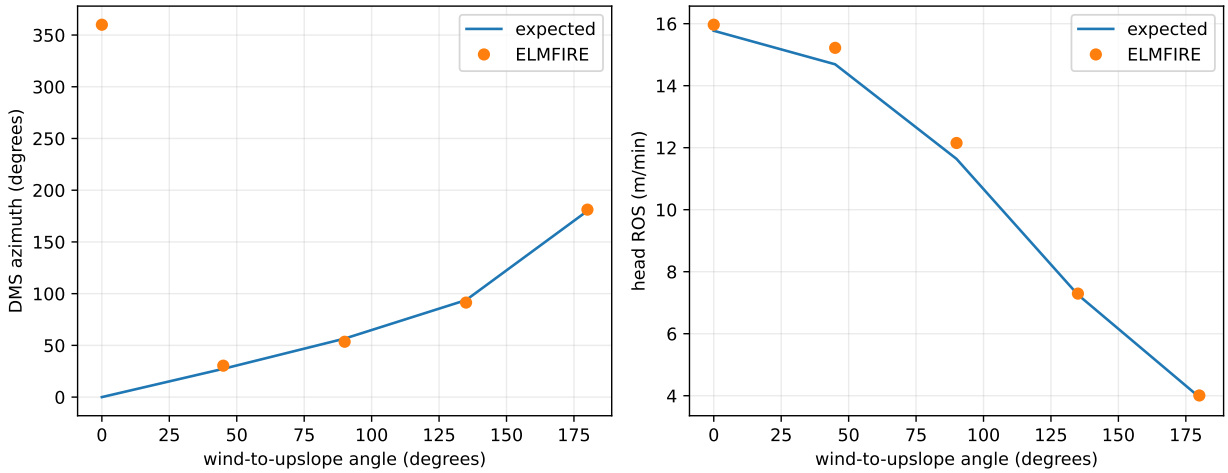


Figure 2: Expected and measured DMS azimuth and head ROS.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum DMS direction error	≤ 5.0	3.05989 degrees	PASS
maximum head-ROS relative error	≤ 0.05	0.0441104 fraction	PASS
maximum length/width relative error	≤ 0.1	0.0961658 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.01156e-07 fraction	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A 180-degree error indicates a wind-from or aspect reversal. A 90-degree error suggests swapped axes or azimuth convention. Correct direction but excessive head speed suggests scalar addition. Correct DMS and head speed with wrong length/width isolates effective-wind or ellipse construction.

9 Scope and limitations

This test uses one fuel, one slope magnitude, one wind magnitude, uniform terrain, and a single grid. It verifies deterministic code behavior rather than observed fire shape. Complex terrain and resolution convergence remain separate concerns.